👩‍🏫 Staff Guide · Ag Science 1 · Washington Crops: Potatoes, Wheat & Apples (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Mr. Torretta's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
Bedlington Farm PowerPoint	Reading parts 1-2; notes section A
WA Wheat, Barley and Potatoes	Reading part 3; notes section B table
Potato Lab	Notes section A (plant parts)
Cosmic Crisp Apples; Apple Taste Lab	Reading parts 4-5; notes section C taste table
Unit review	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Mr. Torretta's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Mr. Torretta.

🥔 Washington Crops: Potatoes, Wheat & Apples: Guided Notes

Ag Science 1 · Unit 2 · Level 3 (closer to the class work) · goes with Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab

Name: _____

Use the class slides and your reading. Key words:

seed potato | *tuber* | *eye* | *irrigate* | *soft white wheat* | *barley* | *variety* | *orchard* | *grafting* | *pollinate*

A. Bedlington Farms and seed potatoes

1. Bedlington Farms started in _____.
2. A potato is a _____, a swollen underground stem.
3. New sprouts grow from the _____.
4. Why do potato farmers rotate crops?

_____

_____

B. Wheat, barley and the Columbia Basin

5. Fill in the table.

Crop	Where in WA	Main use
Potatoes	Columbia Basin	
Soft white wheat		noodles, crackers, cakes
Barley	eastern WA	

6. Columbia Basin farms _____ because there is little rain.

C. Apples and Cosmic Crisp

7. Washington grows more than _____ of U.S. apples.
8. _____ keeps each apple variety the same.
9. Fill in the table.

Apple	Taste	My rating (1-5)
Gala		
Granny Smith		
Cosmic Crisp		

🥔 Washington Crops: Potatoes, Wheat & Apples: Reading

Ag Science 1 · Unit 2 · Level 3 (closer to the class work) · goes with Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab

Name: _____

Three Washington Crops

Dick Bedlington Farms started in 1942 with 4 acres and now grows about 1,200 acres of potatoes in Deming, Everson, Lynden and Ferndale. They grow certified seed potatoes, potatoes planted to grow new plants. Whatcom is good for seed because it is isolated between the ocean and mountains, so there are fewer insects and diseases, and the soil drains well. They rotate potatoes with wheat, grass, corn and green beans.

A potato is a tuber, the swollen underground stem of the plant. The eyes are buds where new sprouts grow. Above ground the plant has leaves, stems, flowers and small fruit. Farmers harvest when the vines die back.

In the Columbia Basin, long sunny days and rich soil give high potato yields. There is very little rain, so farms irrigate with water from the Columbia River. Eastern Washington's dry hills grow wheat, mostly soft white wheat used for noodles, crackers and cakes. Most Washington wheat is exported, much of it to Asia. Barley is used for animal feed and malt.

Washington grows more than half of the apples in the United States. Bees pollinate the blossoms, and growers thin extra apples so the rest grow big. Grafting, joining a branch of one tree onto the roots of another, keeps each variety the same.

Cosmic Crisp was bred at Washington State University by crossing Enterprise and Honeycrisp. Dr. Bruce Barritt developed it, and growers' fees paid for the research. It stays crisp in storage for about a year. In the apple taste lab, students compare varieties like Gala, Granny Smith, Fuji and Cosmic Crisp.

Check your reading

1. Why is Whatcom County a good place to grow seed potatoes?

2. What kind of wheat does Washington mostly grow, and where does most of it go?

3. What two apples were crossed to make Cosmic Crisp?

🥔 Washington Crops: Potatoes, Wheat & Apples: Quick Check

Ag Science 1 · Unit 2 · Level 3 (closer to the class work) · goes with Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. A potato is a…

A. Tuber (underground stem) | B. Root only | C. Fruit

2. Most Washington wheat is…

A. Corn | B. Used only in Washington | C. Soft white wheat that is exported

3. Grafting is used to…

A. Water trees | B. Kill weeds | C. Keep an apple variety the same

4. Cosmic Crisp was bred at…

A. A grocery store | B. Washington State University | C. Bedlington Farms

5. Why do Columbia Basin farms need irrigation?

6. How do bees help apple growers?

🥔 Washington Crops: Potatoes, Wheat & Apples: Guided Notes · ANSWER KEY

Ag Science 1 · Unit 2 · Level 3 (closer to the class work) · goes with Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab

Name: _____

Use the class slides and your reading. Key words:

seed potato · *tuber* · *eye* · *irrigate* · *soft white wheat* · *barley* · *variety* · *orchard* · *grafting* · *pollinate*

A. Bedlington Farms and seed potatoes

1. Bedlington Farms started in **1942**.
2. A potato is a **tuber**, a swollen underground stem.
3. New sprouts grow from the **eyes**.
4. Why do potato farmers rotate crops?
   To keep soil healthy and stop diseases and pests from building up.

B. Wheat, barley and the Columbia Basin

5. Fill in the table.

Crop	Where in WA	Main use
Potatoes	Columbia Basin	**food (fries, chips, seed)**
Soft white wheat	**eastern WA dryland hills**	noodles, crackers, cakes
Barley	eastern WA	**animal feed and malt**

6. Columbia Basin farms **irrigate** because there is little rain.

C. Apples and Cosmic Crisp

7. Washington grows more than **half** of U.S. apples.
8. **Grafting** keeps each apple variety the same.
9. Fill in the table.

Apple	Taste	My rating (1-5)
Gala	**sweet, mild**	**(student choice)**
Granny Smith	**tart, firm**	**(student choice)**
Cosmic Crisp	**sweet-tart, very crisp**	**(student choice)**

🔑 *Key written by our team from Mr. Torretta's materials, please check.*

🥔 Washington Crops: Potatoes, Wheat & Apples: Reading · ANSWER KEY

Ag Science 1 · Unit 2 · Level 3 (closer to the class work) · goes with Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab

Name: _____

Three Washington Crops

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In the Columbia Basin, long sunny days and rich soil give high potato yields. There is very little rain, so farms irrigate with water from the Columbia River. Eastern Washington's dry hills grow wheat, mostly soft white wheat used for noodles, crackers and cakes. Most Washington wheat is exported, much of it to Asia. Barley is used for animal feed and malt.

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Check your reading

1. Why is Whatcom County a good place to grow seed potatoes?
It is isolated (fewer insects and diseases) and has rich, well-drained soil.

2. What kind of wheat does Washington mostly grow, and where does most of it go?
Soft white wheat; most is exported, much to Asia.

3. What two apples were crossed to make Cosmic Crisp?
Enterprise and Honeycrisp.

🥔 Washington Crops: Potatoes, Wheat & Apples: Quick Check · ANSWER KEY

Ag Science 1 · Unit 2 · Level 3 (closer to the class work) · goes with Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. A potato is a…

- **A. Tuber (underground stem)**
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2. Most Washington wheat is…

- A. Corn
- B. Used only in Washington
- **C. Soft white wheat that is exported**

3. Grafting is used to…

- A. Water trees
- B. Kill weeds
- **C. Keep an apple variety the same**

4. Cosmic Crisp was bred at…

- A. A grocery store
- **B. Washington State University**
- C. Bedlington Farms

5. Why do Columbia Basin farms need irrigation?

There is very little rain; water comes from the Columbia River.

6. How do bees help apple growers?

They pollinate the blossoms so apples can form.

🔑 *Key written by our team from Mr. Torretta's quizzes and notes, please check.*